Chapter 1 Introduction to State Pricing

In the subsequent chapters, the same ideas will re-appear.

Rmk. Basic setting: finite-state, one-period, state pricing is also related to beta models. □

1A Arbitrage and State Prices

$\{1, \cdots, S\}$ states, N securities, $(D_{ij})_{N\times S}$ gives the payoff matrix.

$q \in \mathbb{R}^N$ is the price vector, $\theta \in \mathbb{R}^N$ is the portfolio vector □

Def. Arbitrage: a portfolio θ s.t. $q\cdot\theta \le 0$, $D^T\theta > 0$

or $q\cdot\theta < 0$ $D^T\theta \ge 0$ □

Def. A state-price vector $\psi \in \mathbb{R}^S_{++}$ s.t. $q = D\psi$ □

$\boxed{\text{Thm.}}$ There is no arbitrage iff there is a state-price vector

proof. Let $L = \mathbb{R}\times\mathbb{R}^S$, $M = \{(-q^T\theta, D^T\theta) : \theta \in \mathbb{R}^N\}$ be a linear subspace of L

Let $K = \mathbb{R}_+ \times \mathbb{R}^S_+$ a cone in L.

[1] K and M are closed and convex. there is no arbitrage if $K\cap M = \{0\}$

[2] $\Rightarrow$ If $K\cap M = \{0\}$, separating hyperplane theorem tells us $\exists$ linear functional $F: L \to \mathbb{R}$ such that

$$F(z) < F(x) \quad \forall z \in M, x \in K \quad x \ne 0$$

so $\exists\, \alpha > 0$, $\psi \gg 0$ in $\mathbb{R}^S$ $F(v, c) = \alpha v + \psi\cdot c$

$$-\alpha q\cdot\theta + \psi\cdot(D^T\theta) = 0 \Rightarrow \psi/\alpha \text{ is the state-price vector}$$

[3] $\Leftarrow$ If $\exists\psi, \forall\theta$ $q\cdot\theta = \psi^T D^T\theta$, when $D^T\theta \ge 0$. $q\cdot\theta \ge 0$

$D^T\theta > 0$ $q\cdot\theta > 0$

□

1B. Risk-Neutral Prob

View any $p \in \mathbb{R}^S$ with $p_1 + \cdots + p_S = 1$ as a vector of prob. of the corresponding states. Given ψ for dividend-price (D, q)

Let $\psi_0 = \psi_1 + \cdots + \psi_S$ and $\hat{\psi}_j = \psi_j / \psi_0$

Then we have $(\hat{\psi}_1, \cdots, \hat{\psi}_S)$, $\forall i \in \{1, \cdots, N\}$

$$\frac{q_i}{\psi_0} = \hat{E}(D_i) \equiv \sum_{j=1}^{S} \hat{\psi}_j D_{ij}$$

1C Optimality and Asset Pricing

An <u>agent</u> is defined by a strictly increasing utility func. $U: \mathbb{R}^S_+ \to \mathbb{R}$ and an <u>endowment</u> $e \in \mathbb{R}^S_+$,

Let $X(q, e) := \{e + D^T\theta \in \mathbb{R}^S_+ : \theta \in \mathbb{R}^N, q \cdot \theta \leq 0\}$

She solves $\sup_{c \in X(q,e)} U(c)$ (1)

$\square$

prop. If $\exists$ solution to (1), then there is no arbitrage

If U is continuous and $\exists$ no arbitrage, then $\exists$ sol. to (1)

$\square$

Thm. Suppose c^* is a strictly positive sol. to (1), that U is continuously differentiable at c^*, $\partial U(c^*)$ of partial derivatives is strictly positive, then $\exists \lambda > 0$ such that $\lambda \partial U(c^*)$ is a state-price vector

proof. $\forall \theta \in \mathbb{R}^N$, $\exists$ scalar $K > 0$ such that $c^* + \alpha D^T\theta \geq 0$ $\forall \alpha \in [-K, K]$ because c^* is strictly positive.

Let $g_\theta : [-K, K] \to \mathbb{R}$: $g_\theta(\alpha) = U(c^* + \alpha D^T\theta)$

Suppose $q \cdot \theta = 0$, optimality $\Rightarrow$ $g'(0) = \partial U(c^*)^T D^T\theta = 0$

$\forall \theta \in \mathbb{R}^N$, $q \cdot \theta = 0$, then $\partial U(c^*) D^T\theta = 0$

$\exists$ scalar μ s.t. $\partial U(c^*) D^T = \mu q^T$,

$D^T\theta_0 > 0 \Rightarrow q \cdot \theta^0 > 0 \Rightarrow \mu > 0$

Hence $q = \frac{1}{\mu} D \partial U'(c^*)$ $\quad$ $\frac{1}{\mu}\partial U'(c^*)$ is the state-price vec

$\square$